MIL-F-63051A (AR)
1 July 1985
SUPERSEDING
MIL-F-63051 (AR)
18 January 1977

MILITARY SPECIFICATION

FUZE, M934 SERIES, ACTUATORS FOR

This specification is approved for use by U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, test, manufacture and acceptance requirements for the arming and abort actuators to be used in the M934 Series Fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FEDERAL

TT-I-1795 - Ink, Marking Stencil, Opaque for Porous Surface

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	- Fuze and Fuze Components, Environmental Performance Tests for
MIL-STD-794	- Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810C	- Environmental Test Methods (Revision Level C to be used)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9282665	- Actuator, Arming
9282666	- Actuator, Abort

(Copies of specifications, standards, drawings and publications required by the manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NAVAL ORDNANCE DOCUMENT

NAVORD REPORT 2101 - Statistical Methods Appropriate for Evaluation of Fuze Explosive-Train Safety and Reliability

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CODE OF FEDERAL REGULATIONS

49 CFR 170-190 - Department of Transportation Rules for Regulations for the Transportation of Explosive and Other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. Orders for the above publication should cite "49 CFR 0-199 (latest revisions)".)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and processes. Materials and processes shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements of the applicable drawings, specifications and standards.

3.3 Ambient conditions. Unless otherwise specified, all performance characteristics shall be demonstrated under the following ambient conditions:

a. Temperature Range: +65°F to +95°F

b. Relative Humidity: 10 to 95 percent

c. Barometric Pressure: Local average ±2 inches of mercury

3.4 Performance

3.4.1 Solderability. The component lead wires shall meet the requirements of MIL-STD-202, Method 208.

3.4.2 Terminal strength. Lead wires and terminals shall be permanently secured internally and externally, as applicable, to assure that the normal movement of the lead wires will not result in damage, wear or excessive strain to the component. The component, after being tested for terminal strength, shall show no evidence of loose, broken, deformed or displaced parts and shall meet the all-fire functional requirements of 3.4.6 and the exhibited force requirement of 3.4.7.

3.4.3 Lead-to-lead resistance. The DC resistance between the lead wires shall be as specified in the detailed drawing.

3.4.4 Insulation resistance

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3.4.4.1 Pre-fired condition. The resistance between the lead wires shorted together and the case shall be 50 megohms minimum at all temperatures.

3.4.4.2 Post-fired condition. The resistance between the lead wires shorted together and the case shall be 100 ohms minimum at 140°F.

3.4.5 No-fire characteristic. No piston actuator shall function when the continuous current specified in the detailed drawing is applied to the terminals.

3.4.6 All-fire characteristic. All piston actuators shall function when the continuous minimum current is applied to the terminals for the maximum time, both limits as specified in the detailed drawing.

3.4.7 Exhibited force. When initiated, the piston actuator shall be capable of moving a specified force through a specified distance in less than specified maximum time. The limits are specified in the applicable detail drawing.

3.4.8 Environmental. There shall be no evidence of mechanical or electrical damage during or after exposure to the following environmental conditions.

3.4.8.1 Thermal shock

3.4.8.2 Mechanical shock

3.4.8.3 Sinusoidal vibration

3.4.8.4 Random vibration

3.4.8.5 Acceleration

3.4.9 Rough handling. It is not required that the components be operable after these tests. The criteria by which the samples are judged to have withstood the following tests are that (1) no element shall explode and (2) no parts shall be broken, deformed, displaced or come apart in such a manner as to make the assembly unsafe to handle or dangerous to use as determined by examination.

3.4.9.1 Jolt

3.4.9.2 Jumble

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3.4.10 Bruceton analysis. When tested and calculated according to the Bruceton Method, the all-fire current shall be equal to or greater than the minimum specified in the detailed drawing at a 95% confidence level. The no-fire current shall be equal to or less than the maximum no-fire current specified in the detailed drawing, at a 95% confidence level.

3.4.11 Identification of piston actuator explosives. The identity and quantity of all explosive materials used in the component shall be furnished along with the components when submitted for acceptance. No change of material or quantity from that specified in the detailed drawing shall be made without approval of the government procuring activity.

3.4.12 Marking. Identification marking shall be in accordance with MIL-STD-130.

3.4.13 First article inspection. This specification contains technical provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

3.4.14 Workmanship. The component shall be fabricated and finished in a thorough, workmanlike manner. It shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance program. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made part of this detailed specification.

4.2 Classification of inspection. In conformance with MIL-A-48078, the following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer. The First Article Sample shall consist of the following items in the quantities listed:

PART NUMBER	TITLE	QUANTITY
9282665	Actuator, Arming	340
9282666	Actuator, Abort	340

4.3.2 Inspections to be performed. Examination shall be in accordance with MIL-A-48078 and Table I specified herein.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63051A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 6		DRAWING NUMBER 9282665/9282666
	Actuator, Arming Actuator, Abort				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Lead-to-Lead Resistance (Pre-fire)		100%	3.4.3	4.5.3
	Insulation Resistance (Pre-fire)		100%	3.4.4.1	4.5.4
	Examination for Defects	30		3.2	4.4.2.1

NOTES:

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63051A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Actuator, Arming Actuator, Abort	2 OF 6	9282665/9282666
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	1/ Temperature (^{o}F + 5°F)			2/	3/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	GROUP I	-40	Amb	+140	0	30		
	Solderability		30				3.4.1	4.5.1
	Terminal Strength		30				3.4.2	4.5.2
	All Fire Characteristic/ Exhibited Force 4/		30				3.4.6, 3.4.7	4.5.6, 4.5.7
	GROUP II	-40	Amb	+140	0	30		
	Lead-to-Lead Resistance (Pre-Fire)	10	10	10			3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)	10	10	10			3.4.4.1	4.5.4
	GROUP III Bruceton Analysis	-40	Amb	+140	0	45		
	Step Side Determination (All-Fire)	5					3.4.10	4.5.15
	Test (All-Fire)	-40					3.4.10, 3.4.6	4.5.15, 4.5.6
	GROUP IV Bruceton Analysis	-40	Amb	+140	0	45		
	Step Size Determination (No-Fire)			5			3.4.10	4.5.15
	Test (No-Fire)			40			3.4.10, 3.4.5	4.5.15, 4.5.5

1/ Test Temp and test quantities required for individual exam or test except as otherwise required by test procedure para. For exposure type test, use temp specified in Test Method. 2/ Number of defectives permitted in the test group. 3/ Total quan of sample components required for test group. 4/ The All Fire characteristic shall be tested after the Terminal Strength Test (Minor 202)

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63051A (AR)

PARAGRAPH	TITLE	SHEET 3 OF 6	DRAWING NUMBER
	Actuator, Arming Actuator, Abort		9282665/9282666 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	1/ Temperature (°F + 5°F)			2/	3/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	GROUP V	-40	Amb	+140	0	20		
	No-Fire Characteristic	5	5	10			3.4.5	4.5.5
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	GROUP VI	40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Shock, Thermal		20				3.4.8	4.5.8 (Visual)
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	All Fire Characteristic Exhibited Force		20				3.4.6, 3.4.7	4.5.6, 4.5.7
	Insulation Resistance (Post-Fire)			20			3.4.4.2	4.5.4
	Lead-to-Lead Resistance (Post-Fire) 4/		20				3.4.3	4.5.16
	GROUP VII	-40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Shock, Mechanical		20				3.4.8	4.5.9 (Visual)

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63051A (AR)

PARAGRAPH	TITLE Actuator, Arming Actuator, Abort				SHEET 4 OF 6		DRAWING NUMBER 9282665/9282666 NEXT HIGHER ASSEMBLY	
CATEGORY	EXAMINATION OR TEST	1/ Temperature (°F + 5°F)			2/	3/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	All Fire Characteristic (Exhibited Force)		20				3.4.6, 3.4.7	4.5.6, 4.5.7
	Insulation Resistance (Post-Fire)			20			3.4.4.2	4.5.4
	Lead-to-Lead Resistance (Post-Fire) 4/		20				3.4.3	4.5.16
	GROUP VIII	-40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Sinusoidal Vibration		20				3.4.8	4.5.10 (Visual)
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	All-Fire Characteristic/ Exhibited Force		20				3.4.6, 3.4.7	4.5.6, 4.5.7
	Insulation Resistance (Post-Fire)			20			3.4.4.2	4.5.4
	Lead-to-Lead Resistance (Post-Fire) 4/		20				3.4.3	4.5.16

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63051A (AR)

PARAGRAPH	TITLE Actuator, Arming Actuator, Abort				SHEET 5 OF 6			DRAWING NUMBER 9282665/9282666 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	1/ Temperature (°F + 5°F)			2/	3/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	GROUP IX	-40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Random Vibration		20				3.4.8	4.5.11 (Visual)
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	All Fire Characteristic (Exhibited Force)		20				3.4.6, 3.4.7	4.5.6, 4.5.7
	Insulation Resistance (Post-Fire)			20			3.4.4.2	4.5.4
	Lead-to-Lead Resistance (Post-Fire) 4/		20				3.4.3	4.5.16
	GROUP X	-40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Acceleration		20				3.4.8	4.5.12 (Visual)
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63051A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Actuator, Arming Actuator, Abort	6 OF 6	9282665/9282666 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	1/ Temperature (°F + 5°F)			2/	3/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	All-Fire Characteristic/ Exhibited Force		20				3.4.6, 3.4.7	4.5.6, 4.5.7
	Pin Protrusion/Retraction (Post-Fire)		20				3.2	Gage
	Insulation Resistance (Post-Fire)			20			3.4.4.2	4.5.46
	Lead-to-Lead Resistance (Post-Fire) 4/		20				3.4.3	4.5.16
	GROUP XI	40	Amb	+140	0	20		
	Lead-to-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Jolt		20				3.4.9	4.5.13 (Visual)
	GROUP XII	-40	Amb	+140	0	20		
	Lead-Lead Resistance (Pre-Fire)		20				3.4.3	4.5.3
	Insulation Resistance (Pre-Fire)		20				3.4.4.1	4.5.4
	Jumble		20				3.4.9	4.5.14 (Visual)

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4.3.3 Rejection. Rejection shall be in accordance with MIL-A-48078.

4.4 Quality conformance assurance

4.4.1 Inspection lot formation. Inspection lot formation shall be in accordance with MIL-A-48078. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot.

4.4.2 Examination. Examination shall be in accordance with MIL-A-48078. Unless otherwise specified sampling shall be in accordance with MIL-STD-105, Inspection Level II. AQL levels, number of sample units to be inspected for inspections requiring fixed samples, and requirements for 100% inspection are specified in 4.4.2 and subparagraphs, Quality Conformance Inspection. All defects shall be major unless otherwise specified.

a. Lot inspection procedure. Inspection of the components and assemblies designated in 4.4.2.1 shall be performed in accordance with 4.4.2.

b. Lot test procedure. Tests on the asemblies designated in 4.4.2.2 shall be performed in accordance with 4.4.2.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Actuator, Arming Actuator, Abort		9282665/9282666
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Case Split or Cracked		100%	3.2	Visual
MAJOR					
101	Lead-to-Lead Resistance (Pre-Fire)		100%	3.4.3	4.5.3
102	Insulation Resistance (Pre-Fire)		100%	3.4.4.1	4.5.4
103	Pin Protrusion (Pre-Fire) (9282665)		0.40%	3.2	Gage
104	Pin Protrusion (Pre-Fire) (9282666) 1/		0.40%	3.2	Visual
MINOR					
201	Pin Diameter (9282665 only)		0.65%	3.2	Gage
202	Body Diameter		0.65%	3.2	Gage
203	Thread Diameter (Major) (9282665)		0.65%	3.2	Gage
204	Thread Diameter (Minor) (9282665)		0.65%	3.2	Gage
205	Gold Plating, Leads		0.65%	3.2	Gage
206	Marking		0.65%	3.4.12	Visual
207	Identification of Explosives 2/			3.4.11	Certificate of Compliance
208	Workmanship		0.65%	3.4.14	Visual

NOTES:
1/ The pin shall not protrude through the end seal.
2/ Each lot of explosives will be separately identifiable with one or more lots of actuators.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63051A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Actuator, Arming Actuator, Abort	1 OF 1	9282665/9282666
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Environmental Exposure Sequence	1/	0.65%	3.4.8	
MAJOR					
101	Mechanical Shock (Groups A&B) 2/		0.65%	3.4.8.2	4.5.9 (Visual)
102	Sinusoidal Vibration (Groups A&B) 2/		0.65%	3.4.8.3	4.5.10 (Visual)
103	Random Vibration (Groups A&B) 2/		0.65%	3.4.8.4	4.5.11 (Visual)
104	Acceleration (Groups A&B) 2/		0.65%	3.4.8.5	4.5.12 (Visual)
105	Lead-to-Lead Resistance (Pre-Fire) 2/		0.65%	3.4.3	4.5.3 (Ambient)
106	Insulation Reistance (Pre-Fire) 2/		0.65%	3.4.4	4.5.4 (Ambient)
107	No-Fire Characteristic (+140°F)		0.65%	3.4.5	4.5.5
108	All-Fire Characteristic/Exhibited Force (-50°F)			3.4.6 3.4.7	4.5.6, 4.5.7
109	Pin Protrusion/Retraction (Post-Fire)		0.65%	3.2	Gage
110	Insulation Resistance (Post-Fire)		0.65%	3.4.4.2	4.5.4 (+140°F)
111	Lead-to-Lead Resistance (Post-Fire) 3/		0.65%	3.4.3	4.5.16 (Ambient)

NOTES: 1/ Number of sample units per 4.4.2/ sample exposed to all tests listed.
2/ Samples divided into 2 equal groups, A&B. "A" units tested at +140°F; "B" units tested at -50°F.
3/ Abort actuator 9282666 only.

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4.4.3 Testing. The tests (First Article and Quality Conformance) listed in the item detail specification shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract.

4.5 Methods of inspection.

4.5.1 Solderability. This test shall be performed on units subjected to the all-fire test, 4.5.6. There shall be no special preparation of terminals prior to testing. Testing shall be in accordance with MIL-STD-202, Test Method 208.

4.5.2 Terminal strength. The piston actuator shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A, except that a 2 pounds minimum pull force shall be applied for a period of 30 seconds per lead.

4.5.3 Lead-to-lead resistance. This test shall be performed in accordance with MIL-STD-202, Test Method 303, using a current of 1.1 milliamperes maximum.

4.5.4 Insulation resistance. The insulation resistance test, pre-fired condition, shall be performed in accordance with MIL-STD-202, Test Method 302, Test Condition B. The insulation resistance test, post-fired condition, shall be performed using a test voltage of 28VDC. The test potential shall be applied etween the lead wires shorted together and the case.

4.5.5 Maximum no-fire current. The component shall be tested by applying the specific maximum current (+0, -10 mA) through the unshorted lead wires for the designated period, both parameters as specified in the detailed drawing.

4.5.6 Minimum all-fire Current. The component shall be tested by applying the specified minimum current (-0, +50 mA) through the unshorted lead wires for the designated period, both parameters as specified in the detailed drawing.

WARNING

Unloaded extendable pins may eject when actuator is initiated.

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4.5.7 Exhibited force. The piston shall be mounted perpendicular to the orientation in which gravity would aid in passing the test. The load shall be an opposing force of the magnitude specified in the detail drawing.

4.5.8 Thermal shock. The component shall be tested in accordance with MIL-STD-202, Test Method 107, Test Condition A except that the high temperature shall be +162°F +5°F in lieu of 85°C +3°C.

4.5.9 Mechanical shock. This test shall be performed in accordance with MIL-STD-202, Test Method 213, Test Condition A, except the peak value of the shock pulse shall be 150g. The component shall be mounted rigidly in a manner which permits testing after the environmental exposure test. Each component shall receive the shock pulses, one each along the following axes:

a. Longitudinal axis of actuator, vertical, with the leads up.

b. Longitudinal axis of actuator, vertical, with the leads down.

c. Longitudinal axis of actuator horizontal.

4.5.10 Sinusoidal vibration. This test shall be performed in accordance with MIL-STD-331, Test Method 119, Procedure 2 (2 hours per axis - ambient temperature), except the test level may be 0.4 inches double amplitude or 5g, whichever is less. The test temperatures shall be per the First Article and Quality Conformance tables herein.

4.5.11 Random vibration. This test shall be performed in accordance with MIL-STD 810C, Method 514.2, Procedure V, Part 2. Curve AG and Note 3 apply.

4.5.12 Acceleration. This test shall be performed in accordance with MIL-STD-202, Method 212, Test Condition A, at 100g.

4.5.13 Jolt. This test shall be performed in accordance with MIL-STD 331, Test 101.1, except the actuator shall be packed for shipment per 5.2 while subjected to the test.

4.5.14 Jumble. This test shall be performed in accordance with MIL-STD-331, Test 102.1 except the actuators shall be packed for shipment while subjected to the test.

4.5.15 Bruceton analysis. The minimum all-fire current and the maximum no-fire current shall be determined using constant

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current testing. They shall be determined through the use of Bruceton Analysis testing as specified in NAVORD Report 2101 using the input Current (I) as the independent variable. The piston actuator output force and operating time as specified in the detailed drawing shall be measured and used as the criteria for success or failure of the individual operation.

4.5.15.1 All-fire. The minimum all-fire current shall be computed from the test data as follows:

$$I_{AF} = \bar{I}_R$$

$\bar{I}_R$ = 50% reliable all-fire current

σ_R = standard deviation of all-fire current

4.5.15.2 No-fire. The maximum no-fire current shall be computed from the test data as follows:

$$I_{NF} = \bar{I}_U$$

$\bar{I}_U$ = 50% reliable no-fire current

σ_U = standard deviation of no-fire current

4.5.16 Lead to lead resistance. This test shall be performed in accordance with MIL-STD-202, Test Method 303, only subsequent to 25 milliseconds after the test specimen has been fired.

5. PACKAGING

5.1 Preservation - packaging and packing. Preservation - packaging and packing and identification shall be in accordance with 49CFR 171-190 to afford protection from damage during shipment.

5.1.1 Level C. The level of protection shall be Level C in accordance with MIL-STD-794. A suitable number of packaged (inner) actuators shall be packaged in an outer box in accordance with 49CFR 171-190.

5.1.2 Shielding. Actuators, in groups of 5, or a portion thereof, shall be wrapped in a layer of conductive foil. The leads of each component shall be shorted together.

5.2 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129. Each inner container and intermediate container shall be printed, stamped or labelled in accordance with the following information:

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a. Quantity, component name, component part number.

b. Lot Number, month and year (date packed).

c. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4-inch high applied with black waterproof ink. Stencil ink shall conform to TT-I-1795.

d. Exterior boxes shall be marked with DOT designation. The number of actuators packed in one (1) outer pack shall be shown in the marking.

6. NOTES

6.1 Intended Use. The actuators covered by this specification are intended for use in the M934 Series Fuze for the Stinger Missile.

6.2 Ordering Data. See MIL-A-48078.

6.3 Submission of Designs for Approval. See MIL-A-48078. Submit equipment designs, as required, to U.S. Army Armament Research and Development Center, ATTN: AMSMC-QAR-I (D), Dover, New Jersey, 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR:

Preparing Activity:
Army-AR:
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